Arduino Programming Cheat Sheet

University of Asia Pacific
Electrical & Electronics Engineering

Structure & Flow

Basic Program Structure

void **setup()** {
//Runs once when sketch starts
}
void **loop()** {
//Runs repeatedly
}

Comments

//this is a single line
/*this is a multiline*/

Setup

pinMode(pin,[INPUT | OUTPUT | INPUT_PULLUP]);

/* Sets the mode of the digital I/O.
it can be set as an input, output, or an input with an internal pull-up resistor. */

Control Sturcture

if(condition)
{
//if condition is TRUE, do something.
}
else
{
// otherwise, do this.
}

for(initialization; condition; increment)
{
// do this.
}
/* The 'for' statement is used to repeat a block of statements enclosed in curly braces. An increment counter is usually used to increment & terminate the loop*/
Ex:
for(int x = 0; x < 43; x++){
print(x);// prints 0 to 42
}

Built-in Functions

Digital I/O

Pins: 0 - 13 A0 - A5
pinMode(pin, (INPUT | OUTPUT | INPUT_PULLUP))

digitalWrite(pin, val);
/* val = High or Low write a HIGH or a LOW value to a digital pin. */
int var = digitalRead(pin);
/* Reads the value from a specified digital pin, either HIGH or LOW. */

Analog I/O

Pins: A0 - A5
analogWrite(pin, val);
/* Writes an analog value to a pin.
val = integer value from 0 - 255. */

PWM Out - pins 3,5,6,9,10,11

int var = analogRead(pin);
/* Reads the value from a specified analog pin, In Range of (0 - 1023) . */

Data Type: Variables, Arrays, & Data

void // nothing is returned
boolean // 0, 1, false, true
char // 8 bits: ASCII character
byte // 8 bits: 0 to 255, unsigned
int // 16 bits: 32,768 to 32,767, signed.
long /* 32 bits: 2,147,483,648 to 2,147,483,647, signed.*
float // 32 bits, signed decimal

Operators

Constans:

HIGH | LOW
INPUT | OUTPUT
true | false

Mathematica Operators:

= (assignment)
+ (addition)
- (subtraction)
* (multipication)
/ (division)
% (modulus)

Logical Operators:

== (boolean equal to)
!= (not equal to)
< (less than)
> (greater than)
<= (less than or equal to)
>= (greater than or equal to)
&& (Boolean AND)
|| (Boolean OR)
! (Boolean NOT)

Bitwise Operators:

& (bitwise AND)
| (bitwise OR)
^ (bitwise XOR)
~ (bitwise INVERT)
var « n (bitwise shift left by n bits)
var » n (bitwise shift right by n bits)

Arduino Library

Serial - communication via UART

begin(long speed);
end();
int **available**() //num, of bytes available
print(data); **println**(data);

Wire.h - I2C communication

begin(); //join a master
begin(addr); //join a slave

Servo.h - control servo motor

attach(pin, min_us, max_us);
write(angle); //0 to 180
writeMicroseconds(useconds);//1000 - 2000; 1500 is midpoint
int **read**(); //0 to 180 angle
bool **attached**(); detach();

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